LESSON PLAN

WORKSHOP PRACTICES (PRACTICAL) (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Guided missiles sub-assemblies, airframe, propulsion unit, guidance unit, control unit, warhead unit etc; Demonstration in weapon demo room.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Guided missiles sub-assemblies, airframe, propulsion unit, guidance unit, control unit, warhead unit etc; Demonstration in weapon demo room.
5. Specific Objectives	23.1.1.1 Trainee will be able the basic knowledge of air craft armaments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of guided missile sub-assemblies in weapon demo room; Safety precautions regarding live weapon handling; Instructor supervision during handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary components of a guided missile?
(b) Define the function of the propulsion unit in a missile.
(c) List three types of warheads used in air craft armaments.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Missile Airframe Structure • Propulsion Unit Components • Guidance Unit Systems • Control Unit Mechanisms • Warhead Unit Types	L & Ds	PPT, IB	15 min
(b)	• Weapon Demo Room Safety Protocols • Handling of Propulsion Units • Inspection of Guidance Systems • Control Unit Operation • Warhead Safety Mechanisms	L & Ds	PPT, PDF	20 min
(c)	• Integration of Sub-assemblies • System Interactions • Maintenance Requirements • Storage and Transport • Final Assembly Check	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the guidance unit directs the missile to its target.
(b) Describe the role of the control unit in stabilizing flight.
(c) Identify the safety precautions required when handling warhead units.

12. Home Assignment and Follow Up	Prepare a brief note on the basic principles of propulsion systems used in guided weapons to be discussed in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	The basic principle of propulsion system used in guided weapons, present day propulsive systems and propellants for rockets and their classifications. Practical on propulsion system.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min